Distributed Consensus and The Raft Protocol

Page 1: Architecture and State Machine Replication

Distributed systems require consensus algorithms to ensure replicated state machines remain consistent despite

Nodes in a Raft cluster exist in one of three states: Follower, Candidate, or Leader. In normal operation, there is

Distributed Consensus and The Raft Protocol

Page 2: Leader Election Mechanisms and Heartbeats

Followers transition to the Candidate state when their randomized election timer expires without receiving heartb

A Candidate initiates an election by incrementing its current term and sending RequestVote Remote Procedure C

Distributed Consensus and The Raft Protocol

Page 3: Log Replication and Split-Brain Safety

When a Leader receives a command from a client, it appends the command to its local log as an uncommitted e

Split-brain protection is guaranteed because any quorum majority that elects a new Leader must overlap with the